Practical-8

Input-




GNU nano 8.7.1 memory.c *
printf("malloc completed\n");
// calloc
b = (int *)calloc(5, sizeof(int));
if (b == NULL) {
printf("calloc failed\n");
free(a);
return 1;
}
printf("calloc completed\n");
// realloc
temp = (int *)realloc(a, 10 * sizeof(int));
if (temp == NULL) {
printf("realloc failed\n");
free(a);
free(b);
return 1;
}
a = temp;
for (int i = 5; i < 10; i++)
a[i] = i + 1;
printf("realloc completed\n");
// Free allocated memory
free(a);
free(b);
printf("Memory freed successfully\n");
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

Nano cow.c



```
 GNU nano 8.7.1                          cow.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

#define SIZE 1000000

int main() {
    int *data = malloc(SIZE * sizeof(int));

    if (data == NULL) {
        perror("malloc");
        return 1;
    }

    for (int i = 0; i < SIZE; i++)
        data[i] = i;

    printf("Parent PID: %d\n", getpid());
    printf("Memory allocated and initialized.\n");

    pid_t pid = fork();

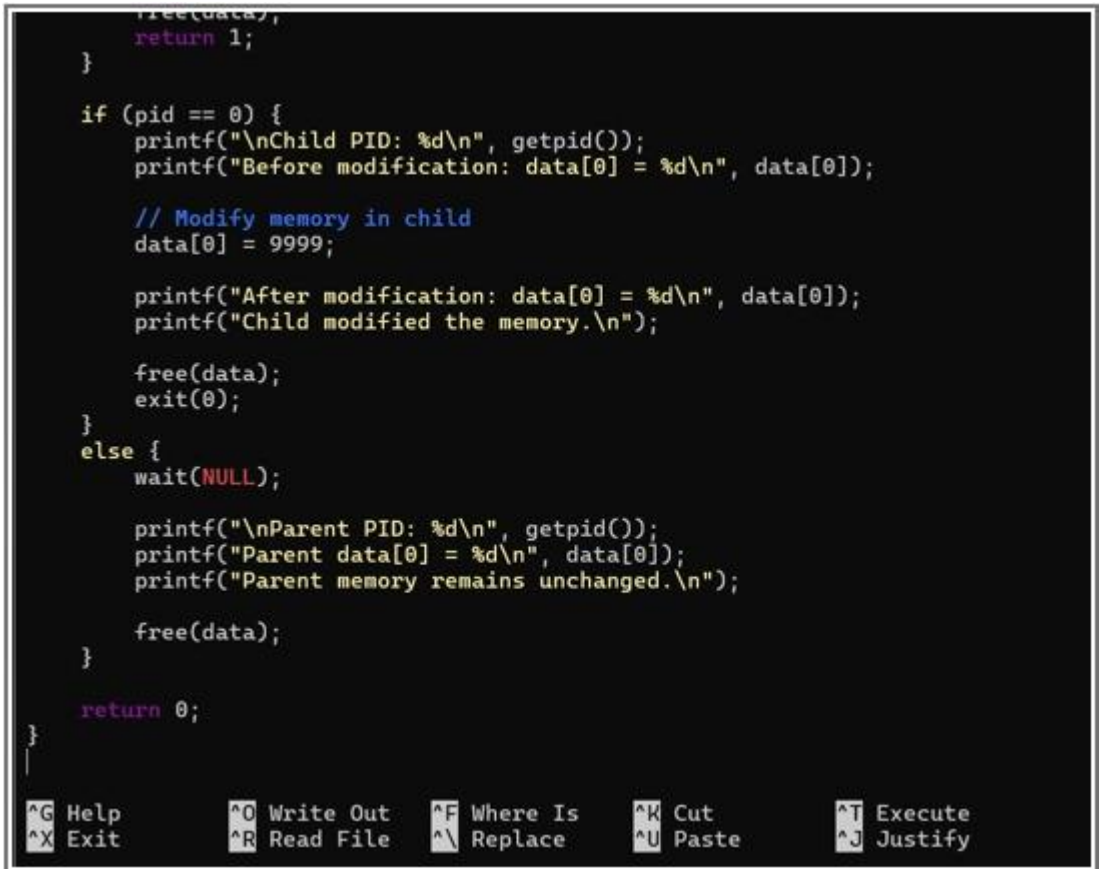
    if (pid < 0) {
        perror("fork");
        free(data);
        return 1;
    }

    if (pid == 0) {
        printf("\nChild PID: %d\n", getpid());
        printf("Before modification: data[0] = %d\n", data[0]);

        // Modify memory in child
        data[0] = 9999;

        printf("After modification: data[0] = %d\n", data[0]);

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```



```
        free(data);
        return 1;
    }

    if (pid == 0) {
        printf("\nChild PID: %d\n", getpid());
        printf("Before modification: data[0] = %d\n", data[0]);

        // Modify memory in child
        data[0] = 9999;

        printf("After modification: data[0] = %d\n", data[0]);
        printf("Child modified the memory.\n");

        free(data);
        exit(0);
    }
    else {
        wait(NULL);

        printf("\nParent PID: %d\n", getpid());
        printf("Parent data[0] = %d\n", data[0]);
        printf("Parent memory remains unchanged.\n");

        free(data);
    }

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

Output-

```
keathana@Chandu:~$ mkdir Pra-8
keathana@Chandu:~$ cd pra-8
-bash: cd: pra-8: No such file or directory
keathana@Chandu:~$ cd Pra-8
keathana@Chandu:~/Pra-8$ nano memory.c
keathana@Chandu:~/Pra-8$ gcc memory.c -o memory
keathana@Chandu:~/Pra-8$ ./memory
malloc completed
calloc completed
realloc completed
Memory freed successfully
keathana@Chandu:~/Pra-8$ valgrind --version
valgrind-3.26.0
keathana@Chandu:~/Pra-8$ sudo apt update
[sudo: Authentication Password:
sudo: Authentication failed, try again.
[sudo: Authentiction Password:
sudo: Authentication failed, try again.
[sudo: Authenticaion Password:
sudo: maximum 3 incorrect authentication attempts
keathana@Chandu:~/Pra-8$ sudo apt update
[sudo: Authentication Password:
sudo: Authentication failed, try again.
[sudo: Authentiction Password:
sudo: Authentication failed, try again.
[sudo: Authenticaion Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
[674 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Compone
nts [46.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Component
s [97.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packa
ges [302 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Com
ponents [43.4 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Comp
```

```
s [97.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packa
ges [302 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Com
ponents [43.4 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Comp
onents [190 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Co
mponents [1056 B]
Fetched 1766 kB in 3s (595 kB/s)
132 packages can be upgraded. Run 'apt list --upgradable' to see them.
keathana@Chandu:~/Pra-8$ sudo apt install valgrind
valgrind is already the newest version (1:3.26.0-0ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 132
keathana@Chandu:~/Pra-8$ valgrind --leak-check=full --show-leak-k
ind=all ./memory
==919== Memcheck, a memory error detector
==919== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==919== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==919== Command: ./memory
==919==
malloc completed
calloc completed
realloc completed
Memory freed successfully
==919==
==919== HEAP SUMMARY:
==919==     in use at exit: 0 bytes in 0 blocks
==919==   total heap usage: 4 allocs, 4 frees,  1,104 bytes allocated
==919==
==919== All heap blocks were freed -- no leaks are possible
==919==
==919== For lists of detected and suppressed errors, rerun with: -s
==919== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
keathana@Chandu:~/Pra-8$ nano cow.c
keathana@Chandu:~/Pra-8$ gcc cow.c cow
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find cow: No such file or directory
collect2: error: ld returned 1 exit status
keathana@Chandu:~/Pra-8$ gcc cow.c -o cow
keathana@Chandu:~/Pra-8$ ./cow
```

```
keathana@Chandu:~/Pra-8$ ./cow
Parent PID: 983
Memory allocated and initialized.

Child PID: 984
Before modification: data[0] = 0
After modification: data[0] = 9999
Child modified the memory.

Parent PID: 983
Parent data[0] = 0
Parent memory remains unchanged.
keathana@Chandu:~/Pra-8$ ./cow 8
[1] 996
Memory allocated and initialized.

Child PID: 1001
Before modification: data[0] = 0
After modification: data[0] = 9999
Child modified the memory.

Parent PID: 996
Parent data[0] = 0
Parent memory remains unchanged.
keathana@Chandu:~/Pra-8$ ps aux | grep cow
 keathana    1008   0.0  0.0   4128   2428 pts/0    S+   06:23   0:00 grep --co
lor=cow
keathana@Chandu:~/Pra-8$ cat /proc/<PID>/status
-bash: PID: No such file or directory
keathana@Chandu:~/Pra-8$ ls
cow  cow.c  memory  memory.c
keathana@Chandu:~/Pra-8$
```